MEDIUM

GEOMETRY AND TRIGONOMETRY

Lines, angles, and triangles

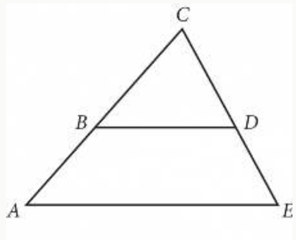
02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62°, what is the measure of angle CAE ?

A. 58°

B. 60°

C. 62°

D. 120°

Q2

GRID-IN GEOMETRY AND TRIGONOMETRY

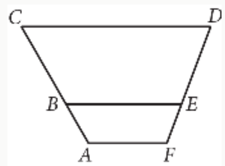
Triangles *ABC* and *DEF* are congruent, where *A* corresponds to *D*, and *B* and *E* are right angles. The measure of angle *A* is 69°. What is the measure, in degrees, of angle *F*?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 GEOMETRY AND TRIGONOMETRY



In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

A. 16.8

B. 17.5

C. 18.4

D. 19.6

Q4

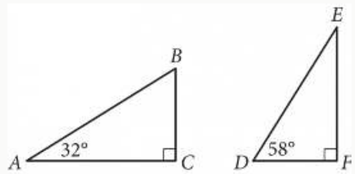
In triangle ABC, the measure of angle A is 50°. If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

A. 50°

B. 65°

C. 80°

D. 100°



Triangles ABC and DEF are shown above. Which of the following is equal to the ratio $\frac{BC}{AB}$?

A. $\frac{DE}{DF}$

B. $\frac{DF}{DE}$

C. $\frac{DF}{EF}$

D. $\frac{EF}{DE}$

Q6

GEOMETRY AND TRIGONOMETRY

Two nearby trees are perpendicular to the ground, which is flat. One of these trees is 10 feet tall and has a shadow that is 5 feet long. At the same time, the shadow of the other tree is 2 feet long. How tall, in feet, is the other tree?

A. 3

B. 4

C. 8

D. 27

Q7

GEOMETRY AND TRIGONOMETRY

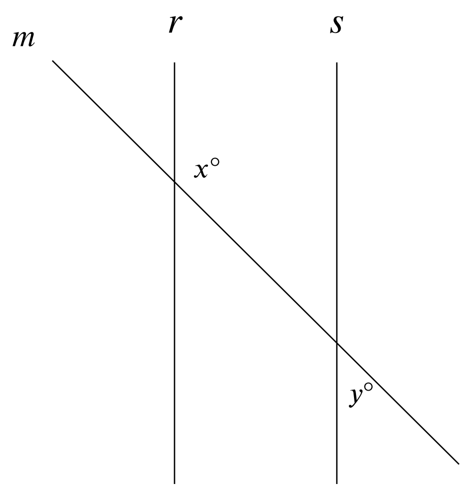
In triangle ABC, the measure of angle A is $54°$, the measure of angle B is $90°$, and the measure of angle C is $\frac{k}{2}°$. What is the value of k?

A. 36

B. 45

C. 72

D. 108



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled m, r, and s.
- Line m intersects both line r and line s.
- At the intersection of line m and line r, 1 angle is labeled clockwise from top left as follows:
    - Top right: x°
- At the intersection of line m and line s, 1 angle is labeled clockwise from top left as follows:
    - Bottom right: y°
- A note indicates the figure is not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

A. $x < 115$

B. $x > 115$

C. $x + y < 180$

D. $x + y > 180$

Q9 GEOMETRY AND TRIGONOMETRY

Quadrilateral $P^{'}Q^{'}R^{'}S^{'}$ is similar to quadrilateral $PQRS$, where P, Q, R, and S correspond to $P^{'}$, $Q^{'}$, $R^{'}$, and $S^{'}$, respectively. The measure of angle P is $30°$, the measure of angle Q is $50°$, and the measure of angle R is $70°$. The length of each side of $P^{'}Q^{'}R^{'}S^{'}$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle $P^{'}$?

A. $10°$

B. $30°$

C. $40°$

D. $90°$

Q10

GEOMETRY AND TRIGONOMETRY

Triangle ABC is similar to triangle XYZ, such that A, B, and C correspond to X, Y, and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC. The measure of side AB is 16. What is the measure of side XY?

A. 14

B. 16

C. 18

D. 32